Math 133: Quiz 7

Friday, October 30, 2026 10 points

Name: _____

Show your work in Problems 1–4 and justify your classifications. Give exact answers; derivatives may be left in factored form. A simple scientific calculator is allowed, but is not needed.

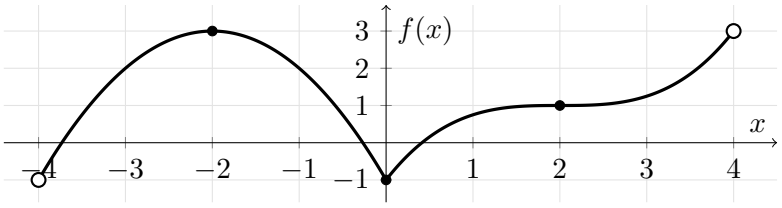
1. (2 points) Find all critical points of $v(x) = xe^{2x}$. Give their x-coordinates. You do not need to classify them.

Critical numbers: _____

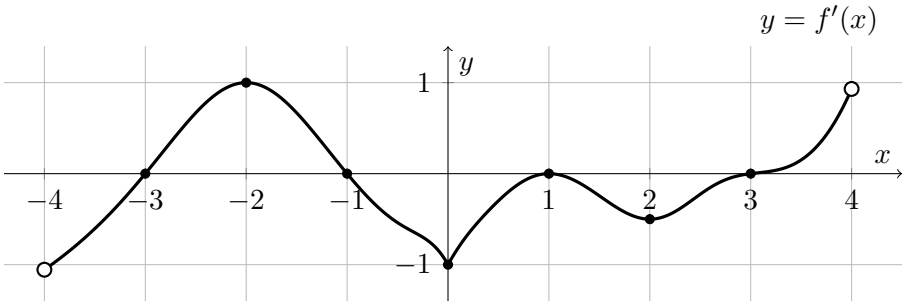
2. (3 points) Find all critical points of $u(x) = 4x^3 + 9x^2 - 12x + 5$. Give their x-coordinates, and classify each as a local maximum, a local minimum, a saddle point, or neither.

Critical numbers: _____

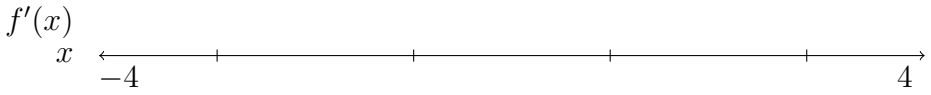
3. (2 points) The graph below shows f. List all critical points of f in $(-4, 4)$ by their x-coordinates, and classify each as a local maximum, a local minimum, a saddle point, or neither.



4. (2 points) For this problem, f is differentiable on $(-4, 4)$. The graph below shows f'.



(a) Make a sign chart for f' on $(-4, 4)$.



(b) List all critical points of f in $(-4, 4)$ by their x-coordinates, and classify each as a local maximum, a local minimum, a saddle point, or neither.

5. (1 point) Assume f is continuous but not necessarily differentiable. Circle True or False for each statement.

(a) Every critical point is a local maximum or a local minimum. **True** **False**

(b) If f has a local extremum at an interior domain point c, then $f'(c) = 0$. **True** **False**

(c) If f is differentiable near c and f' changes from negative to positive at c, then f has a local minimum at c. **True** **False**

(d) If $f'(c)$ does not exist, then c must be a critical number of f. **True** **False**